Landing on the Wrong Altimeter Setting: A Pre-registered Study of Business Jets and Airline Jets in Open ADS-B Data

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Abstract

An aircraft that lands with the wrong altimeter setting flies its barometric altitudes at the wrong height with no barometric indication of it. Two European studies have measured how often this happens; none has measured it in the United States or compared types of operator. Modern aircraft broadcast the pressure setting selected on board. This pre-registered study compared that broadcast setting, for about 714,000 landings at 119 US and European airports on 30 days in 2026, with the consensus of other aircraft landing at the same airport within 30 minutes. The first hypothesis was supported: business jets landed with a setting at least 2 hPa from the consensus more often than airline jets at the same airport on the same day (Mantel-Haenszel odds ratio 3.4, 95% interval 2.9 to 4.0). The second, that landing with the standard setting still selected is more frequent in Europe, was not supported. Independent review then found that the pre-registered measure, the last value broadcast before landing, picks up a spurious switch to the standard setting in the final seconds on some European airliners. With a corrected measure the first result is unchanged (odds ratio 3.5) and the second disappears (rate ratio 0.95). On the corrected measure the setting was wrong by 2 hPa or more on 2.0 per 1,000 landings, by 5 hPa or more on 0.20, and by 10 hPa or more (about 280 ft) on 0.06. A second pre-registered study then tested the corrected measure on 30 further days (about 730,000 landings). The business-jet result replicated (odds ratio 4.0, 3.5 to 4.7), as did the rates (2.0, 0.19 and 0.06 per 1,000). In both samples the consensus was within 1 hPa of the airport weather report for 97% of landings, and using the weather report as the reference gave an odds ratio of 2.6 to 2.8. Most of the difference between business jets and airline jets consists of small discrepancies of 2 to 3 hPa. The difference is larger for errors of 5 hPa or more (odds ratio 8.6, 5.2 to 14.2), 13.9 (8.9 to 21.8) in the second sample, where it rests on few events and on aircraft seen only a few times. It falls to about 2 among the most frequently seen aircraft. Which altimeter the broadcast reflects is not defined, so on some business jets a wrong broadcast may not mean the crew flew on a wrong setting.

1 Introduction

On 23 May 2022 an Airbus A320 on a satellite-based approach to Paris-Charles de Gaulle descended to about six feet above the ground about 0.9 NM short of the runway. The controller had passed an altimeter setting 10 hPa too high, the crew set it, and every barometric indication on board looked normal [1]. The European Union Aviation Safety Agency has issued and revised a safety bulletin on incorrect altimeter settings [2].

How often aircraft land on a wrong setting has been measured twice, both times in Europe. The French civil aviation safety directorate compared the setting downlinked to radar with

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the setting displayed to controllers for 1,694,266 landings at six French airports from 2019 to 2023. It found a difference of 2 hPa or more at the runway threshold on 2,269 landings (about 1.3 per 1,000) and of 10 hPa or more on 387 (about 0.23 per 1,000). About 27% of the inconsistencies had the standard setting on board. The rate at Le Bourget, a business-aviation airport, was three times the rate at Charles de Gaulle, which the report attributed to "very different fleets" without further analysis [3]. A EUROCONTROL study built a detection algorithm on broadcast data and, as reported by France to an ICAO regional group, found 196 errors above 5 hPa in 747,353 flights at 378 European airports [4, 5]. The full text of that paper could not be obtained for this study.

No published rate was found for the United States, and no study was found that compares business aviation with airlines or one region with another. The United States differs from Europe in two relevant ways. Settings are given in inches of mercury, and the change from the standard setting to the local one is made at 18,000 ft, well before the approach, where in Europe it is made at a few thousand feet [3, 6].

This study uses open data to measure the rate at 119 airports in both regions and to test two pre-registered hypotheses: that business jets land on a wrong setting more often than airline jets at the same airport on the same day (H1), and that landing with the standard setting still selected is more frequent at European than at US airports (H2).

2 Methods

The hypotheses, measures and analysis were committed to a public repository before any altimeter-setting data from the study days was read (branch `prereg/08-wrong-altimeter-setting`, commit `be41aa0`, 11 November 2026, 13:18 UTC). Section 3 lists departures from that plan.

Data. The adsb.lol archive publishes each day's ADS-B traces under an open licence [7]. The study uses 30 UTC days in 2026, drawn at random for an earlier study of approach paths that never used the altimeter-setting field. Airports (119, chosen in advance, 76 in the United States and 43 in Europe) and aircraft types (65 business-jet and 58 airline jet types) were fixed lists. Airline-type aircraft were kept only with an airline-format callsign. Airport elevations came from OurAirports [8].

Setting at landing, as pre-registered. A landing is the last airborne point of a track aligned with a runway and followed by ground movement or no later climb. The pre-registered setting at landing is the last value of the selected barometric pressure setting recorded for the aircraft in the 120 seconds before that point [9]. Values outside 940 to 1060 hPa were dropped.

Reference. The reference for a landing is the median setting at landing of other aircraft landing at the same airport within 30 minutes either side. It required at least five other aircraft, at least 60% of them within 1.0 hPa of their median. Landings without a reference were excluded.

Errors. The difference is the setting at landing minus the reference. Thresholds were 2 hPa (primary), 5 hPa and 10 hPa. "Standard left set" means a setting within 0.5 hPa of 1013.25 that differed from the reference by 2 hPa or more. Two exclusions were made without regard to class: settings matching station pressure at airports above 300 ft (use of QFE), and all landings by any aircraft that was wrong on at least half of four or more landings.

Analysis. H1: Mantel-Haenszel common odds ratio of an error of 2 hPa or more, business jet against airline jet, across strata of airport and UTC date, with a Cochran-Mantel-Haenszel test, a Robins-Breslow-Greenland interval and an interval from resampling whole airports.

H2: the ratio of the rate of "standard left set" at European airports to that at US airports, among landings where the reference was at least 3 hPa from standard, with a p-value from permuting the region label across airports. Holm correction across the two tests. The method was built on one pilot day outside the sample, using pooled results and a run with shuffled labels; no result split by class or region was computed until all 30 days were processed.

Replication and weather-report check. After the first sample was analysed and reviewed, a second pre-registration fixed the corrected measure of Section 3 and four hypotheses (commit `7306d3b`, 11 November 2026, 14:39 UTC), and 30 further days were drawn at random from the 263 eligible days of 2026 not already used. Section 6 reports that study. It also compares the reference with the most recent routine or special weather report (METAR) at each airport, taken from the Iowa State University archive [10].

3 Deviations from the pre-registration

The frozen code was run unchanged and its hashes match the pre-registration. Section 4 reports its results as registered.

A flaw in the registered measure, found in review. Two independent reviewers found that the recorded field mixes two sources. One is the ADS-B broadcast at 0.8 hPa resolution; the other has 0.1 hPa resolution and is probably a Mode S register decoded by the receiver software. On some aircraft, mostly European airliners, the second source shows the standard setting for a few seconds around touchdown after the correct setting had been shown throughout the approach. Because the registered measure takes the last value, it counted these as errors. They were about half of the European airline errors and most of the European "standard left set" events. A second problem was a small number of aircraft whose broadcast value jumped between impossible values.

Corrected measure (post hoc). Section 5 and everything after it use a corrected setting at landing: the most frequent value in the same 120-second window, with the landing excluded if three or more different values were recorded in the window. Everything else is as registered. Differences were rounded to 0.1 hPa before thresholds were applied. The corrected setting differed from the registered one on 1,784 landings, and 80 landings were excluded as unstable. All results on the corrected measure are post hoc.

Other. The analyses in Sections 5.4 and 7 were designed after the results were seen. The two rate tables by aircraft type use lower cut-offs than the frozen script. Stratum dates span 31 UTC dates because three landings fell just after midnight.

4 Results as pre-registered

Sample. The 30 days held 809,959 landings: 714,916 by airline jets and 95,043 by business jets. A setting was recorded in the last two minutes on 97.0% of them. After exclusions (27 implausible values, 3 landings on QFE, and 89 landings by 9 aircraft, 8 of them business jets, under the half-wrong rule), 713,835 landings were analysed: 650,307 by 17,617 airline aircraft and 63,528 by 11,618 business jets. Table 1 shows that far fewer business-jet landings could be analysed.

H1. Within the same airport and date, the odds ratio of an error of 2 hPa or more for business jets against airline jets was 3.42 (95% interval 2.94 to 3.98; resampling airports, 2.77 to 4.27; Holm-adjusted $p < .001$; 684 informative strata). The crude rates over all airports were 8.53 per 1,000 (542 of 63,528) and 1.43 per 1,000 (930 of 650,307). H1 is supported.

Table 1: Coverage by class and region.

Group	Landings	Setting recorded	Analysed
Airline jets, Europe	275,237	99.6%	93.6%
Airline jets, United States	439,679	97.2%	89.3%
Business jets, Europe	17,416	97.9%	74.9%
Business jets, United States	77,627	86.9%	65.0%

H2. Among eligible landings the standard setting was recorded on 71 of 188,811 in Europe (0.38 per 1,000) and 58 of 303,311 in the United States (0.19 per 1,000): rate ratio 1.97 (0.99 to 3.49; permutation $p = .072$). H2 is not supported.

Registered rates. The registered measure gave 2.06 errors of 2 hPa or more per 1,000 landings, 0.26 of 5 hPa or more and 0.083 of 10 hPa or more. These are overstated in Europe for the reason given in Section 3 and are superseded by Table 2.

5 Results on the corrected measure

Nothing from here on is confirmatory.

5.1 How often

On the corrected measure 713,758 landings were analysed. The setting differed from the reference by 2 hPa or more on 1,411 (1.98 per 1,000; 95% interval 1.80 to 2.17), by 5 hPa or more on 144 (0.20 per 1,000; 0.17 to 0.24), and by 10 hPa or more on 43 (0.060 per 1,000; 0.043 to 0.078). Table 2 and Figure 1 give the rates by region and class.

Table 2: Landings per 1,000 with the setting wrong by at least the amount shown (corrected measure). Crude rates; event counts in brackets.

	Landings	2 hPa	5 hPa	10 hPa
Europe, all	270,643	0.33 (89)	0.16 (44)	0.048 (13)
United States, all	443,115	2.98 (1,322)	0.23 (100)	0.068 (30)
Airline jets, Europe	257,594	0.26 (67)	0.10 (27)	0.023 (6)
Airline jets, United States	392,651	2.05 (804)	0.14 (56)	0.038 (15)
Business jets, Europe	13,049	1.69 (22)	1.30 (17)	0.54 (7)
Business jets, United States	50,464	10.27 (518)	0.87 (44)	0.30 (15)

Three things stand out. Errors of 5 hPa or more were somewhat more frequent in the United States than in Europe (0.23 against 0.16 per 1,000). Errors of 2 to 5 hPa were about sixteen times as frequent in the United States (2.76 against 0.17 per 1,000). And the European rates are well below the two earlier European studies: 0.33 and 0.05 per 1,000 at 2 and 10 hPa against about 1.3 and 0.23 in the French study [3], and 0.16 at 5 hPa against about 0.26 reported for EUROCONTROL [5]. The French study used the radar downlink, which may be the same source that produced the spurious values removed here, but this was not tested.

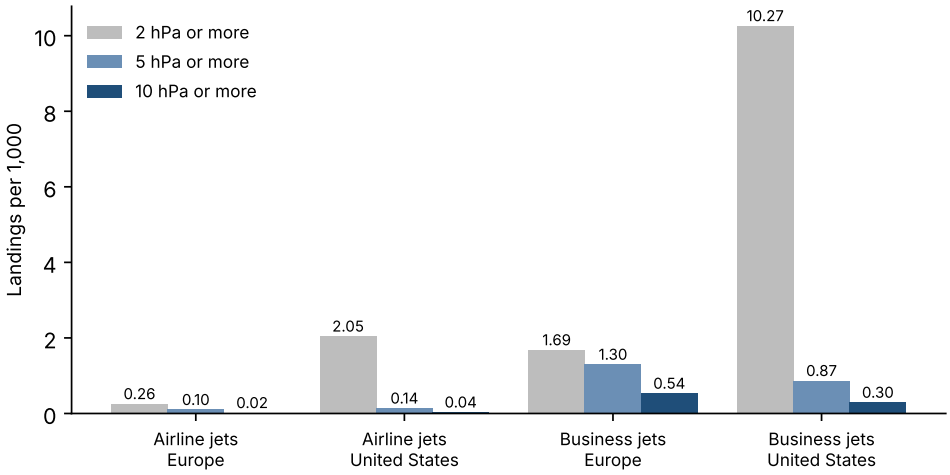


Figure 1: Landings per 1,000 with the setting wrong by at least the amount shown, corrected measure, crude rates.

5.2 Business jets against airline jets

On the corrected measure the H1 odds ratio was 3.46 (2.97 to 4.02; resampling airports, 2.81 to 4.29). It held under the pre-registered sensitivity analyses re-run on the registered measure, which gave between 3.1 and 5.1: excluding periods of changing pressure (4.0), a 15-minute reference window (3.5), aircraft with three or more landings (3.1), the United States alone (3.3), Europe alone (5.1; 2.9 to 9.0), adding the hour to the strata (3.6), ground-confirmed landings only (3.2), and leaving out each airport in turn (3.3 to 3.7). The reviewers confirmed it with conditional and fixed-effects logistic models (3.4 to 3.5), with the reference built from airline aircraft only (3.7), and at airport-dates where at least 90% of landings of both classes could be analysed (3.7).

By size of error. For errors of 2 to 5 hPa the odds ratio was 3.1 (2.7 to 3.7), on 293 business-jet and 777 airline events in informative strata. For 5 hPa or more it was 8.6 (5.2 to 14.2), on 49 and 79 events. For 10 hPa or more it was 16.8 (6.9 to 41.2), on 21 and 20 events. The larger ratios are imprecise, and the strata that inform them differ from threshold to threshold, so they should not be read as a smooth trend.

By how often the aircraft was seen. The gap narrows among aircraft that landed often in the sample (Table 3). Restricted to aircraft with ten or more landings the odds ratio was 2.6 (2.1 to 3.1), and with twenty or more, 2.0 (1.4 to 2.8). Of the 22 business-jet errors of 10 hPa or more, 19 were by aircraft with fewer than ten landings in the sample. Counting one randomly chosen landing per aircraft gave a median odds ratio of 2.7 over 30 draws (range 2.1 to 4.2).

Table 3: Errors of 2 hPa or more per 1,000 landings, by the number of landings the aircraft made in the sample.

Landings by the aircraft	1	2–3	4–9	10–19	20 or more
Business jets	13.5	10.8	10.4	7.2	5.2
Airline jets	16.4	3.0	1.5	1.5	1.3

By airport. The crude business-jet rate rose with the share of business jets at the airport: 3.0 per 1,000 where they were under 5% of traffic, 5.8 at 5 to 20%, 8.5 at 20 to 60% and 18.3 above 60%. Airline rates rose too (1.1, 1.4, 2.7 and 7.6). At business-aviation airports the reference is mostly made of other business jets and is noisier, so the pooled crude rate of 8.5 per 1,000 overstates what a like-for-like comparison shows. Requiring a reference from ten or more aircraft with over 95% agreement gave an odds ratio of 4.9 (3.9 to 6.1).

By operator and type. Among the 60 airline callsign groups with at least 1,500 landings, rates ran from 0 to 14 per 1,000 with a median of 0.5; several large carriers had none. Regional jets were highest among airline types (2.6 to 5.1 per 1,000). Business-jet types with at least 1,000 landings ran from 3.1 to 16.2 per 1,000. The spread between operators is as wide as the gap between the classes.

5.3 Standard setting at landing

On the corrected measure H2 gave a rate ratio of 0.95 (0.52 to 1.60; $p = .85$). The apparent European excess in Section 4 was the artefact.

The registered definition also catches ordinary small differences that happen to fall on 1013. Requiring the setting to be at 1013 and at least 5 hPa from the reference leaves 53 landings (Table 4). These were about half of business-jet errors of 5 hPa or more (32 of 61) and a quarter of airline ones (21 of 83). Within the same airport and date, business jets had far higher odds (32; 13 to 77), on 27 and 19 events.

Table 4: Landings with 1013 selected and the reference at least 5 hPa away.

Group	Eligible landings	Events	Per 1,000
Airline jets, Europe	158,123	15	0.095
Airline jets, United States	218,843	6	0.027
Business jets, Europe	8,040	14	1.74
Business jets, United States	27,437	18	0.66

5.4 Other observations

- **Small errors in the United States.** Of US errors, 74% for airline jets and 86% for business jets had been held unchanged for five minutes or more. Airline small errors bunched in the first ten minutes of each hour (177, against 101 to 127 in the other ten-minute periods), which fits the hourly weather update. Business-jet small errors were flat across the hour (74 to 87).
- **Direction.** Errors of 2 hPa or more were evenly split. Of errors of 5 hPa or more, 50 were set too high, which puts the aircraft lower than indicated, and 94 too low.
- **Pressure far from standard.** On the registered measure, errors of 10 hPa or more rose with distance of the local setting from standard: for airline jets 0.007, 0.05 and 0.24 per 1,000 within 5 hPa, 5 to 15 hPa and beyond 15 hPa; for business jets 0.21, 0.46 and 1.35.
- **Repeat errors.** Errors were rarely repeated by the same aircraft. A reviewer found that 533 of the 542 business-jet errors came from aircraft that broadcast different values on other flights, and that 530 held one steady value through the approach.

- **Paris.** Le Bourget had too few analysable landings (873) to test the French threefold difference.

6 Replication on a second sample

This section is confirmatory. The measure, exclusions, hypotheses and code were fixed before any of the second sample was downloaded, and the frozen script was run once after all 30 days were processed.

Sample. One drawn day (27 March) could not be processed and was replaced by the first reserve day (25 March), as the plan provided. The 30 days held 825,926 landings, of which 730,324 were analysed: 667,394 by airline jets and 62,930 by business jets. Coverage matched the first sample (analysed: airline jets 93.9% in Europe and 89.3% in the United States; business jets 75.1% and 65.7%). The days are new but the aircraft are not: 99% of the landings were by aircraft that also appear in the first sample.

Hypotheses. All four were supported after Holm correction (Table 5). The primary hypothesis also met its pre-set replication criterion, an odds ratio between 2 and 6.

Table 5: Business jets against airline jets within airport and date: first sample (post hoc on the corrected measure) and second sample (pre-registered). Mantel-Haenszel odds ratios with 95% intervals; business-jet and airline events in informative strata in brackets.

Outcome	First sample	Second sample
R1: wrong by 2 hPa or more	3.5 (3.0–4.0) [342, 856]	4.0 (3.5–4.7) [377, 877]
R2: wrong by 5 hPa or more	8.6 (5.2–14.2) [49, 79]	13.9 (8.9–21.8) [64, 63]
R3: R1, aircraft with 10 or more landings	2.6 (2.1–3.1) [130, 797]	2.6 (2.1–3.2) [135, 830]
R4: 1013 at landing, 5 hPa or more away	32 (13–77) [27, 19]	35 (16–78) [34, 18]

Rates. The setting was wrong by 2 hPa or more on 2.04 per 1,000 landings (1.85 to 2.25), by 5 hPa or more on 0.19 (0.16 to 0.23) and by 10 hPa or more on 0.060 (0.043 to 0.079), against 1.98, 0.20 and 0.060 in the first sample. By group, the rates at 2, 5 and 10 hPa were: airline jets in Europe 0.31, 0.11 and 0.04; airline jets in the United States 2.03, 0.10 and 0.03; business jets in Europe 1.98, 1.19 and 0.47; business jets in the United States 11.3, 1.07 and 0.30. Errors of 5 hPa or more were again somewhat more frequent in the United States (0.21 against 0.16 per 1,000), a difference that could be chance (permutation $p = .22$).

Secondary analyses. The odds ratio was 3.4 (2.9 to 4.0) for errors of 2 to 5 hPa and 20 (9 to 44) for 10 hPa or more, on 19 and 21 events. It was 3.9 in the United States and 7.4 (4.3 to 13.0) in Europe, 3.8 with the hour added to the strata, and 4.8 with the strict reference. Among aircraft with twenty or more landings it was 2.2 (1.6 to 3.0). The pattern in Table 3 recurred: business-jet rates fell from 19.9 per 1,000 for aircraft seen once to 6.4 for those seen twenty times or more, against 1.3 for airline aircraft seen that often.

The reference against weather reports. In the second sample the consensus reference was within 1.0 hPa of the latest weather report for 97.5% of 724,718 landings (97.2% in Europe, 97.6% in the United States) and within 1.7 hPa for 99.9%. The first sample gave 97.4% and 99.9% (this check on the first sample was post hoc). The consensus therefore tracks the published setting. With the weather report as the reference in place of the consensus, which brings in about 17,000 business-jet landings that had too little other traffic for a consensus, the odds ratio was 2.8 (2.5 to 3.1) at 2 hPa and 9.4 (6.6 to 13.4) at 5 hPa; in the

first sample 2.6 (2.3 to 3.0) and 9.9 (6.5 to 15.1). This version counts as an error any setting that lags a report issued up to an hour earlier, so it is the more conservative of the two.

A labelling error in the frozen script. In the output of the United States against Europe comparison the two rate labels are transposed. The ratio and p-value are unaffected; the figures above are taken from the per-region rates.

7 What the result does and does not show

What is established. Business jets land with a broadcast altimeter setting that disagrees with other aircraft at the same airport about three to four times as often as airline jets, compared at the same airport on the same day (about two and a half to three times when the airport weather report is the reference). That was confirmed in a second pre-registered sample of new days. It survives correction of the measure, every sensitivity analysis, alternative statistical models, an airline-only reference, and restriction to well-covered airports. It is not produced by a few faulty transponders.

Most of it is small. Nine in ten of the discrepancies behind that result are between 2 and 5 hPa, mostly 2.4 or 3.2 hPa, equivalent to about 65 to 90 ft. In the United States these look like settings that were right when set and not renewed. For airline jets they follow the hourly update. For business jets they do not, which suggests a setting taken earlier or from another station; a setting from a station within 100 NM is legitimate en route. These are discrepancies from the local setting more than errors in the sense of the Paris incident.

The large errors are few. Errors of 5 hPa or more occurred on about 1 business-jet landing per 1,000 against about 0.13 for airline jets. The second sample gave the same rates. The ratio is large and was confirmed, but in each sample it rests on 60 to 70 business-jet events, about half of them with 1013 selected, and at 10 hPa nearly all came from aircraft seen fewer than ten times.

The broadcast may not be the altimeter the crew flew on. The standards define the field as the setting selected by the crew but not which side it comes from [11]. A reviewer with avionics knowledge noted that on some light jets flown by one pilot, the transponder may take its setting from the unused side, and that such aircraft often lack a warning when the two sides disagree. The data are consistent with that reading and cannot exclude it: 1013 at landing was commonest in older and lighter types and absent in several types with integrated flight decks. On this reading some business-jet "standard at landing" events are an unset second altimeter and not a crew flying on the wrong setting. That would still mean no cross-check between the two sides was made.

The United States against Europe. Large errors were at least as frequent in the United States as in Europe. This does not support treating the problem as a European one that the 18,000 ft transition altitude prevents. The comparison is crude: it does not adjust for the mix of traffic, and the European rates here are lower than both earlier studies.

Consequences. A 10 hPa error is about 280 ft. On an approach with ground-based vertical guidance the glidepath is unaffected, but the decision altitude, the altitude check on the glidepath, step-down and missed-approach altitudes are all displaced. On an approach using barometric vertical guidance the whole path is displaced [1]. Radio altimeters, terrain warning using satellite height and controller warnings remain as defences. The study does not know which type of approach was flown.

8 Limitations

- **The measure is not direct.** The recorded value combines two sources that can disagree, and which altimeter either reflects is unknown. The corrected measure was chosen after seeing the problem in the first sample, then fixed in advance for the second.
- **Inferred reference.** The reference is the consensus of other aircraft. A wrong value given to every aircraft would not be detected. It agreed with weather reports within 1 hPa for 97% of landings; Palm Beach was missing from the weather data.
- **Coverage and selection.** Only 65% of US and 75% of European business-jet landings could be analysed, against about 90% for airline jets. The missing landings are aircraft that do not broadcast a setting, which is optional in the United States, and landings with too little other traffic. A reviewer found that business-jet types with lower broadcast coverage had higher error rates among the aircraft that did broadcast, so the direction of any bias is unknown.
- **Exposure.** Airline aircraft averaged 37 landings in the sample and business jets 5.5. The per-landing comparison is dominated by rarely seen business jets, and the gap is smaller among frequently seen aircraft.
- **Lag counted as error.** A crew holding a correct but older value is counted at the 2 hPa threshold.
- **Resolution.** The broadcast has 0.8 hPa steps, so 2 hPa is effectively 2.4 hPa except where the reference falls on a half step.
- **Classes.** The airline class includes cargo and some military and government flights with airline-format callsigns. Operator type is inferred from aircraft type and callsign. Test and training flights are included.
- **No operational context.** Approach type, weather, minima, crew composition and what ATC said are unknown.
- **Sample.** Sixty days in 2026 at 119 airports chosen by judgement. The second sample has new days but the same airports and almost entirely the same aircraft, so it confirms that the result is stable over time and was not produced by choices made after seeing the data. It does not show that the result would hold for other airports or fleets.
- **Prior work not fully read.** The EUROCONTROL paper was seen only as an abstract and through secondary reports.

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Competing interests. The author is the founder of CitadelAero, which builds AI tools for aviation, and is employed as a business-jet pilot.

Disclaimer. The views expressed are the author's own and do not represent those of any employer or regulator. Nothing here is operational, legal or regulatory advice.

Materials. The pre-registration, code, results and audit record for this paper are at `https://github.com/JamesLeCuirot/Research/tree/prereg/08-wrong-altimeter-setting/papers/08-wrong-altimeter-setting`.

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